

operator. You can find out how it operates by going to <https://jenkins-x.io/v3/about/how-it-works/>.

Step 1: Set up GitHub repository for GitOps. Click on <https://github.com/jx3-gitops-repositories/jx3-minikube/generate> based on the *jx3-gitops-repositories/jx3-minikube* template to create the jenkinsX repository.

If this link does not work log in to GitHub first and retry the link.

Step 2: *git clone* the new repository.

Step 3: *cd* into the *git clone* directory.

Step 4: Set *DOMAIN* variable as IP

address of minikube cluster:

```
export DOMAIN="$(minikube ip).nip.io"
```

Step 5: Set the domain requirements of GitOps using *DOMAIN* variable:

```
jx gitops requirements edit --domain $DOMAIN
```

Step 6: Execute *jx secret edit* in the terminal; it will ask for the Docker registry user name and password for managing secrets and configurations.

Type the Docker hub details and ensure the changes are saved.

\$ *jx secret verify* is used to verify that the external secret resources have the required properties populated in the underlying secret storage.

▪ JenkinsX

Step 1: Use this command to download the most recent version:

```
curl -L https://github.com/jenkins-x/jx/releases/download/v3.10.116/jx-linux-amd64.tar.gz | tar xzv
```

```
prarthana@prarthana-Inspiron-3501:~/gitops/gitops$ jx admin operator --setup "git config --global http.sslVerify false"
git clone URL is https://github.com/prarthanaus0597/gitops.git
now verifying we have a valid git username and token so that we can clone the git repository inside kubernetes...
git username is prarthanaus0597 for URL https://github.com/prarthanaus0597/gitops.git and we have a valid password
running command:

/home/prarthana/.jx3/plugins/bin/helm-3.6.2 upgrade --install \
--set url=https://github.com/prarthanaus0597/gitops.git \
--set username=prarthanaus0597 \
--set password=**** \
--set gitInitCommands=git config --global http.sslVerify false --namespace jx-git-operator --create-namespace jxgo jxgh/jx-git-operator

switched to namespace jx-git-operator so that you can start to create or import projects into Jenkins X: https://jenkins-x.io/docs/v3/create-project/
waiting for the Git Operator to be ready in namespace jx-git-operator...
pod jx-git-operator-79c4d5b9bf-kx2p9 has status Ready
the Git Operator is running in pod jx-git-operator-79c4d5b9bf-kx2p9

waiting for boot Job pod with selector app=jx-boot in namespace jx-git-operator...
```

Figure 5: Start GitOps

The screenshot shows the Octant web interface. The left sidebar contains navigation links for Applications, Namespace Overview, Cluster Overview, Jenkins X, JX OPS, RBAC, and Plugins. The main content area is titled 'Overview' and displays two tables of resources.

environments.jenkins.io/v1

Name	Labels	Namespace	Kind	Promotion	Order	Git URL	Git Branch	Age
dev	env:dev gitops.jenkins-x.io/pipeline.namespaces	jx	Development	Never	<not found>	https://github.com/prarthanaus0597/gitops.git	master	17d
production	env:production gitops.jenkins-x.io/pipeline.namespaces	jx-production	Permanent	Manual	500	<not found>	<not found>	17d
staging	env:staging gitops.jenkins-x.io/pipeline.namespaces	jx-staging	Permanent	Auto	100	<not found>	<not found>	17d

externalsecrets.kubernetes-client.io/v1

Name	Labels	Last Sync	status	Age
bucketrepo-config	gitops.jenkins-x.io/pipeline.namespaces	<not found>	<not found>	17d
docker-registry-secret	app:docker-registry chart:docker-registry-2.2.2	<not found>	<not found>	17d
jenkins-maven-settings	gitops.jenkins-x.io/pipeline.namespaces	<not found>	<not found>	17d
jenkins-x-bucketrepo	gitops.jenkins-x.io/pipeline.namespaces jenkins.io/credentials-type=usernamePassword	<not found>	<not found>	17d
jx-basic-auth-httpswd	gitops.jenkins-x.io/pipeline.namespaces	<not found>	<not found>	17d
jx-basic-auth-user-password	gitops.jenkins-x.io/pipeline.namespaces	<not found>	<not found>	17d
jx-local-secrets	gitops.jenkins-x.io/pipeline.namespaces	<not found>	<not found>	17d
lighthouse-tmac-token	app:lighthouse-lighthouse chart:lighthouse-1.13.8	<not found>	<not found>	17d
lighthouse-oauth-token	app:lighthouse-lighthouse chart:lighthouse-1.13.8	<not found>	<not found>	17d

Figure 6: Environments and secret overview in JX UI